

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME:CSA1610 COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.
Assessment Tool Weightage: 40%

QUESTIONS: 05

Total Marks:50

Annexure A

Industry Problem Solving Task

<i>Criteria</i>	Understanding of Industry Problem (2)	Application of Engineering Concepts (2.5)	Solution Design (2.5)	Use of Tools (2)	Analysis (1)	<i>Total(10)</i>
<i>Weightage</i>	20%	25%	25%	20%	10%	<i>100%</i>
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I G.Hima Bindu(192511377) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G.Hima Bindu

Annexure A

Q1 . Dataset: Supermarket Purchases

Customer Visits/Month Avg Spend (₹) Items Bought

C1	10	2000	15
C2	4	5000	8
C3	12	1800	18
C4	3	6000	6
C5	11	2100	16

Question:

Using the above dataset, propose a clustering approach to segment customers and explain how it supports targeted marketing. (BL3 – 7 Marks)

1. Clustering Approach

For the given supermarket dataset, **K-Means clustering** can be used to segment customers based on their purchasing behavior.

The attributes considered are:

- **Visits/Month**
- **Average Spend (₹)**
- **Items Bought**

Since K-Means is a distance-based algorithm, the attributes should first be **normalized** because Average Spend has much larger numerical values than Visits/Month and Items Bought.

2. Dataset

Customer	Visits/Month	Avg Spend (₹)	Items Bought
C1	10	2000	15
C2	4	5000	8
C3	12	1800	18
C4	3	6000	6
C5	11	2100	16

3. Applying K-Means

A suitable choice is **K = 2**, because the customers show two clear purchasing patterns.

Step 1: Normalize the data

Normalization brings all attributes to a comparable scale, preventing Avg Spend from dominating the distance calculation.

A common method is Min-Max normalization:

A common method is Min-Max normalization:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Step 2: Select K = 2

Two customer segments can be identified.

Step 3: Assign customers to clusters

Based on their purchasing patterns, the likely segmentation is:

Cluster Customers Characteristics

Cluster 1 C1, C3, C5 Frequent visits, lower average spend, more items

Cluster 2 C2, C4 Less frequent visits, higher average spend, fewer items

C1, C3 and C5 show very similar behavior, while C2 and C4 also have similar behavior.

4. Interpretation of Customer Segments

Cluster 1 – Frequent/Regular Shoppers

C1, C3 and C5

Characteristics:

- High visits per month
- Lower average spending per visit
- Higher number of items purchased

These customers visit the supermarket frequently and purchase multiple items but have a relatively lower average bill.

Marketing strategies:

- Loyalty points
- Buy-one-get-one offers
- Weekly discounts
- Personalized product recommendations
- Reward programs for frequent visits

Cluster 2 – High-Value Occasional Shoppers

C2 and C4

Characteristics:

- Low visits per month
- High average spending
- Fewer items purchased

These customers visit less frequently but spend significantly more when they shop.

Marketing strategies:

- Premium product recommendations
- High-value coupons
- Personalized offers
- Bulk purchase discounts
- Special weekend or seasonal promotions

5. How Clustering Supports Targeted Marketing

Clustering allows the supermarket to **divide customers into groups with similar purchasing behavior** instead of applying the same marketing strategy to everyone.

For example:

Frequent shoppers → Loyalty rewards and frequent-visit offers

High-value occasional shoppers → Premium offers and high-value discounts

This results in more personalized marketing campaigns.

6. Benefits

Clustering helps the supermarket to:

- Identify different customer segments.
- Personalize marketing campaigns.
- Improve customer retention.
- Increase purchase frequency.
- Increase average customer spending.
- Provide relevant product recommendations.
- Reduce unnecessary marketing costs.

7. Conclusion

K-Means clustering with $K = 2$ is a suitable approach for this dataset after normalization. It can segment customers into **frequent regular shoppers (C1, C3, C5)** and **high-value occasional shoppers (C2, C4)**.

These segments enable the supermarket to design **targeted marketing strategies**, improve customer satisfaction, increase loyalty, and maximize sales.

Q2. Dataset: Credit Card Transactions

Transaction Amount (₹) Location Time (hrs)

T1	200	Chennai	10
T2	5000	Delhi	2
T3	150	Chennai	11
T4	7000	Mumbai	1
T5	180	Chennai	12

Question:

Design a clustering-based solution using this dataset to detect unusual or fraudulent transactions. (BL4 – 7 Marks)

The goal is to group transactions based on their **Amount, Location, and Time** and identify transactions that do not belong to the normal transaction cluster.

1. Select Features

Use the following features:

Transaction	Amount (₹)	Location	Time
T1	200	Chennai	10
T2	5000	Delhi	2
T3	150	Chennai	11
T4	7000	Mumbai	1
T5	180	Chennai	12

Since **Location is categorical**, encode it numerically before applying clustering, for example:

- Chennai → 0
- Delhi → 1
- Mumbai → 2

2. Apply K-Means Clustering

Choose **K = 2**, representing:

- **Cluster 1:** Normal transactions
- **Cluster 2:** Unusual transactions

Because the amount values are much larger than time/location values, **standardize the features** before clustering.

The normal transactions are likely to form a cluster:

- T1: ₹200, Chennai, 10 hrs
- T3: ₹150, Chennai, 11 hrs
- T5: ₹180, Chennai, 12 hrs

The unusual cluster contains:

- T2: ₹5000, Delhi, 2 hrs
- T4: ₹7000, Mumbai, 1 hr

3. Identify Fraudulent Transactions

Transactions that are **far from the normal cluster centroid** can be considered suspicious.

Transaction	Cluster	Observation	Result
T1	Normal	Low amount, Chennai, daytime	Normal
T2	Unusual	High amount, Delhi, 2 AM	Suspicious
T3	Normal	Low amount, Chennai, daytime	Normal
T4	Unusual	Very high amount, Mumbai, 1 AM	Suspicious
T5	Normal	Low amount, Chennai, daytime	Normal

4. Fraud Detection Rule

A transaction can be flagged as potentially fraudulent if:

Distance from the normal cluster centroid > predefined threshold.

For a real-world system, the threshold can be determined using historical legitimate and fraudulent transaction data.

5. Conclusion

Using **K-Means clustering**, the transactions can be divided into normal and unusual groups. **T1, T3, and T5** form the normal cluster, while **T2 and T4** are detected as suspicious because they involve unusually high amounts and occur at very late/early hours in different locations. Thus, clustering can be used as an **unsupervised fraud-detection technique** when labelled fraud data is unavailable.

Q3. Dataset: Telecom Usage

User	Call Duration (min)	Data Usage (GB)	Recharge (₹)
U1	300	5	200
U2	100	2	100
U3	350	6	220
U4	80	1	90
U5	320	5.5	210

Question:

Explain how clustering can be applied to identify churn-prone users and improve customer retention strategies. (BL4 – 7 Marks)

Clustering can be used to group telecom users based on **Call Duration, Data Usage, and Recharge Amount**. It helps identify users with low engagement who may be more likely to leave the service.

1. Dataset

User	Call Duration	Data Usage	Recharge
U1	300 min	5 GB	₹200
U2	100 min	2 GB	₹100
U3	350 min	6 GB	₹220
U4	80 min	1 GB	₹90
U5	320 min	5.5 GB	₹210

2. Apply Clustering

Use **K-Means clustering with K = 2**. Before clustering, standardize the numerical features so that all features contribute fairly.

The users can be grouped as:

Cluster	Users	Characteristics
Cluster 1	U1, U3, U5	High calls, high data usage and high recharge
Cluster 2	U2, U4	Low calls, low data usage and low recharge

3. Identify Churn-Prone Users

U2 and U4 belong to the low-usage cluster. Their low call duration, data usage, and recharge indicate **low customer engagement**, so they can be considered potential churn-prone users.

4. Retention Strategy

The telecom company can target these users with:

- Discounted recharge plans
- Bonus data or free calling
- Personalized offers
- Loyalty rewards
- Notifications about suitable low-cost plans

High-usage users such as **U1, U3, and U5** can be retained using premium plans and loyalty benefits.

5. Conclusion

Thus, clustering provides an **unsupervised method for customer segmentation**. By identifying low-engagement clusters, telecom companies can target churn-prone users with personalized retention strategies, improving customer satisfaction and reducing churn.

Q4 . Dataset: E-commerce Behavior

User Product Views Clicks Purchases

U1	50	20	5
U2	30	10	2
U3	60	25	6
U4	20	8	1
U5	55	22	5

Question:

Suggest a clustering technique to group users and explain how it improves recommendation systems. *(BL3 – 7 Marks)*

Clustering can group users based on their **Product Views, Clicks, and Purchases**. This helps identify users with similar shopping behaviour and provide personalized recommendations.

1. Dataset

User	Product Views	Clicks	Purchases
U1	50	20	5
U2	30	10	2
U3	60	25	6
U4	20	8	1
U5	55	22	5

2. Suggested Technique – K-Means Clustering

K-Means clustering can be used because the features are numerical.

Choose **K = 2** to form two user groups. Before clustering, the features should be **standardized**.

Expected grouping:

Cluster	Users	Behaviour
Cluster 1	U1, U3, U5	High views, clicks and purchases
Cluster 2	U2, U4	Low views, clicks and purchases

3. Improve Recommendation System

After clustering, the system can recommend products based on the behaviour of users in the same cluster.

For example:

- **U1, U3, U5:** Recommend popular and premium products because they show high engagement and purchasing activity.
- **U2, U4:** Recommend affordable and frequently viewed products to encourage purchases.

4. Benefits

- Provides **personalized recommendations**.
- Groups users with **similar interests and behaviour**.
- Improves customer engagement and purchase probability.
- Reduces the need to analyze every user individually.
- Helps businesses design targeted offers.

5. Conclusion

K-Means clustering helps divide e-commerce users into behaviour-based groups. The recommendation system can then use the preferences and purchasing patterns of each cluster to provide **more relevant and personalized product recommendations**, improving customer satisfaction and sales.

Q5. Dataset: Traffic Data

Location	Vehicle Count	Avg Speed (km/h)	Delay (min)
L1	200	30	10
L2	500	15	25
L3	220	28	12
L4	600	10	30
L5	210	32	9

Question:

Design a clustering-based solution to identify traffic congestion zones and explain how it helps in traffic management. *(BL4 – 7 Marks)*

Clustering can be used to group locations based on **Vehicle Count, Average Speed, and Delay**. Locations with high vehicle count, low speed, and high delay can be identified as **congested zones**.

1. Dataset

Location	Vehicle Count	Avg Speed (km/h)	Delay (min)
L1	200	30	10
L2	500	15	25
L3	220	28	12
L4	600	10	30
L5	210	32	9

2. Apply K-Means Clustering

Use **K-Means clustering with K = 2**:

- **Cluster 1:** Low-congestion locations
- **Cluster 2:** High-congestion locations

Before clustering, standardize the features so that vehicle count, speed, and delay contribute appropriately.

3. Expected Clusters

Cluster	Locations	Characteristics
Low Congestion	L1, L3, L5	Lower vehicle count, higher speed, low delay
High Congestion	L2, L4	High vehicle count, low speed, high delay

Therefore, **L2 and L4 are identified as major traffic congestion zones.**

4. Traffic Management

The traffic management system can use these clusters to:

- Adjust traffic signal timings.
- Divert vehicles to alternative routes.
- Deploy traffic police or emergency teams.
- Provide real-time congestion alerts.
- Improve road capacity planning.
- Introduce intelligent traffic control systems.

5. Conclusion

Clustering helps automatically identify traffic patterns without predefined labels. In this dataset, **L2 and L4 form the high-congestion cluster** because they have high vehicle counts, low average speeds, and high delays. This information helps authorities take **targeted actions to reduce congestion and improve traffic flow.**

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world	Good understanding with minor gaps in	Basic understanding; some requirements unclear	Poor understanding; misinterpret

		problem and constraints	requirements		s problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation